

Rubrics for Quiz-1

Question No 1 (5 Marks, 1 mark each, no partial marks, no explanation, just true or false)

Part (i) – 2 points

Question: For which $Y = (y_1, y_2, y_3)$ does $AX = Y$ have a solution?

Scoring:

- **2 points** – Correctly identifies that $y_3 = 0$ (and y_1, y_2 arbitrary). 9From Row 3 of A , the equation $0 = y_3$ must hold for consistency.
- **1 point** – Partially correct, e.g., only says system is consistent for some Y but does not state $y_3 = 0$ explicitly, or gives wrong/extra conditions.
- **0 points** – Incorrect condition

Correct answer:

$$\boxed{y_3 = 0}$$

Triples $(y_1, y_2, 0)$ for any $y_1, y_2 \in \mathbb{R}$.

Part (ii) – 3 points

1. Check consistency – 1 point

Recognizes $y_3 = 0$ so solution exists.

2. Solve reduced system – 1 point

Correctly writes:

$$x_2 + 2x_3 - 5x_5 = 2, \quad x_4 + 4x_5 = 1$$

and identifies free variables x_1, x_3, x_5 .

3. General solution form – 1 point

Writes general solution in parametric form, e.g.:

$$\begin{cases} x_1 = r, \\ x_2 = 2 - 2x_3 + 5x_5, \\ x_3 = s, \\ x_4 = 1 - 4x_5, \\ x_5 = t \end{cases}$$

or in vector form:

$$X = \begin{pmatrix} 0 \\ 2 \\ 0 \\ 1 \\ 0 \end{pmatrix} + r \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} + s \begin{pmatrix} 0 \\ -2 \\ 1 \\ 0 \\ 0 \end{pmatrix} + t \begin{pmatrix} 0 \\ 5 \\ 0 \\ -4 \\ 1 \end{pmatrix}, \quad r, s, t \in \mathbb{R}.$$

Question No 2 (5 Marks, 1 mark each, no partial marks, no explanation, just true or false)

a. True

Theorem 6. If A is an $m \times n$ matrix and $m < n$, then the homogeneous system of linear equations $AX = 0$ has a non-trivial solution.

Proof. Let R be a row-reduced echelon matrix which is row-equivalent to A . Then the systems $AX = 0$ and $RX = 0$ have the same solutions by Theorem 3. If r is the number of non-zero rows in R , then certainly $r \leq m$, and since $m < n$, we have $r < n$. It follows immediately from our remarks above that $AX = 0$ has a non-trivial solution. ■

b. False

(b) [1 pt] The matrix

$$\begin{pmatrix} 0 & 1 & -2 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$$

is in row-reduced echelon form.

3rd column first row element should be 0

c. true

Definition. A vector space (or linear space) consists of the following:

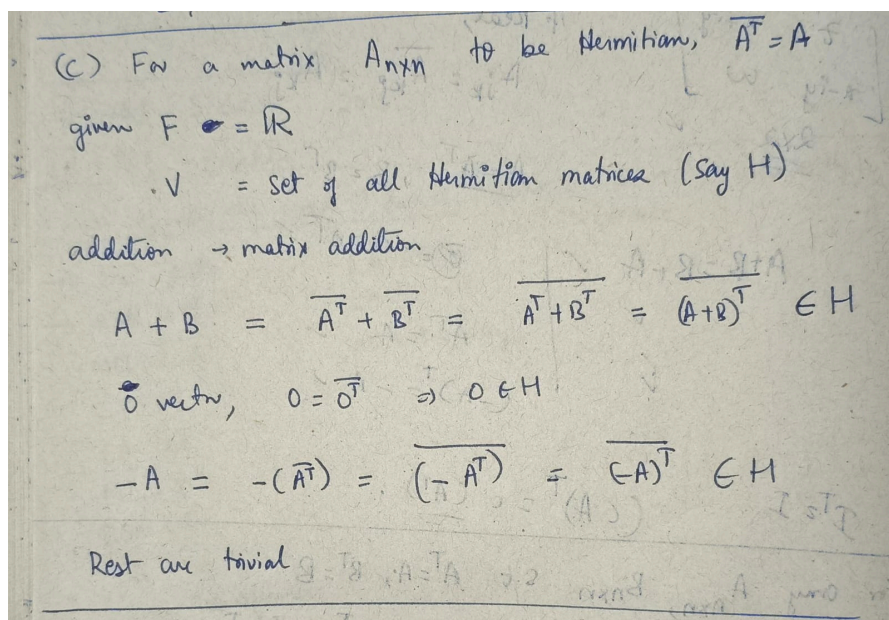
1. a field F of scalars;
2. a set V of objects, called vectors;
3. a rule (or operation), called vector addition, which associates with each pair of vectors α, β in V a vector $\alpha + \beta$ in V , called the sum of α and β , in such a way that
 - (a) addition is commutative, $\alpha + \beta = \beta + \alpha$;
 - (b) addition is associative, $\alpha + (\beta + \gamma) = (\alpha + \beta) + \gamma$;

(c) there is a unique vector 0 in V , called the zero vector, such that $\alpha + 0 = \alpha$ for all α in V ;

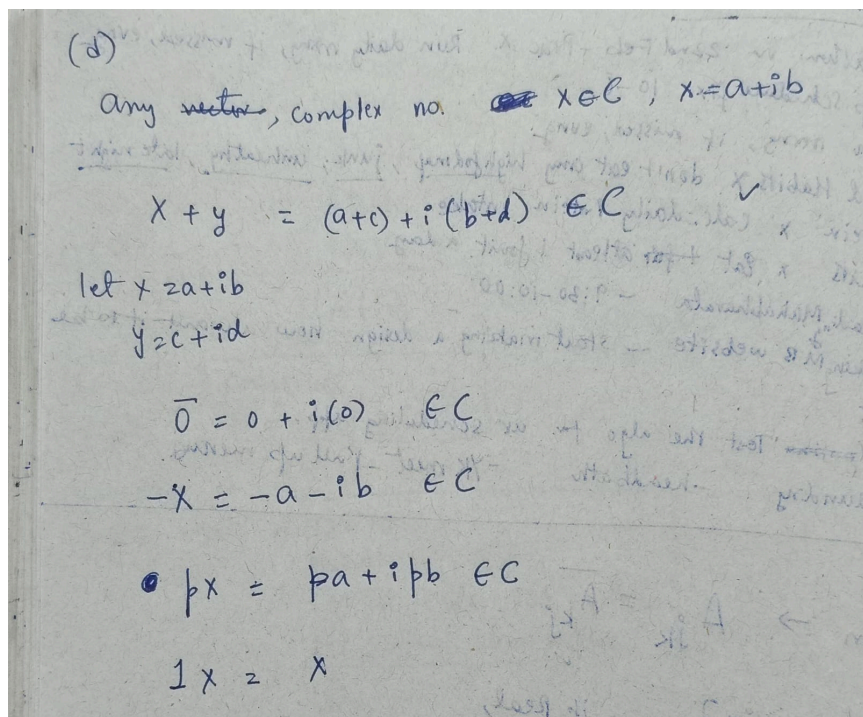
(d) for each vector α in V there is a unique vector $-\alpha$ in V such that $\alpha + (-\alpha) = 0$;

4. a rule (or operation), called scalar multiplication, which associates with each scalar c in F and vector α in V a vector $c\alpha$ in V , called the product of c and α , in such a way that

- (a) $1\alpha = \alpha$ for every α in V ;
- (b) $(c_1c_2)\alpha = c_1(c_2\alpha)$;
- (c) $c(\alpha + \beta) = c\alpha + c\beta$;
- (d) $(c_1 + c_2)\alpha = c_1\alpha + c_2\alpha$.



d. True



e. False

The statement is False because a vector space must always contain the zero vector (0). For the system $AX = Y$ with Y not equal to 0, the zero vector is not a solution since $A(0) = 0 \neq Y$, meaning it fails a fundamental axiom of vector spaces.

Question No 3 (5 Marks)

Solution of Question No 3:

Let A and A' be $m \times n$ matrices and let B be an $n \times m'$ matrix, where m, n, m' are natural numbers.

- (a)
- **1 Point:** The product AB is defined when the number of columns of A equals the number of rows of B . Since A is $m \times n$ and B is $n \times m'$, this condition is satisfied for all natural numbers m, n, m' . Hence, the product AB is always well-defined.
 - **1 Point:** For the product BA , matrix B is $n \times m'$ and matrix A is $m \times n$. The product BA is defined only when $m' = m$.
- (b)
- **1 Point (0.5 each):** From part (a), the only product that is defined for all values of m, n, m' is AB . Hence $C = AB$. Since A is $m \times n$ and B is $n \times m'$, the resulting matrix C has dimensions $m \times m'$. Therefore, C has m rows and m' columns.
 - **2 Points (1 point for each condition):** Now assume $m = m'$ and $m > n$. Let us take

$$B = \begin{bmatrix} \mathbb{I}_n & 0 \end{bmatrix},$$

which is an $n \times m$ matrix. Now let us also consider,

$$A = \begin{bmatrix} \mathbb{I}_n \\ 0 \end{bmatrix}, \quad A' = \begin{bmatrix} \mathbb{I}_n \\ v \end{bmatrix},$$

where v is any nonzero $(m - n) \times n$ matrix. Then $A \neq A'$. Now,

$$BA = \begin{bmatrix} \mathbb{I}_n & 0 \end{bmatrix} \begin{bmatrix} \mathbb{I}_n \\ 0 \end{bmatrix} = \mathbb{I}_n,$$

and similarly,

$$BA' = \begin{bmatrix} \mathbb{I}_n & 0 \end{bmatrix} \begin{bmatrix} \mathbb{I}_n \\ v \end{bmatrix} = \mathbb{I}_n.$$

Thus all required conditions are satisfied.

[If only equations are given rather than an example: -1 point overall]

Question No 4 (5 Marks)

Part (a): Equivalences [1 pt]

1.0 pt: States that (a), (b), and (c) are all equivalent.

0.0 pt: Any other combination.

Part (b): Left Inverse Proof [2 pts]

1.0 pt: Setup: Correctly applies the left-inverse B to both sides of the equation $AX=0$ (i.e., $B(AX) = B(0)$).

0.5 pt: Associativity: Properly groups the matrices $(BA)X = 0$.

0.5 pt: Conclusion: Uses $BA = I$ to show $IX = 0$ implies $X = 0$.

Part (c): Left Inverse to Invertible [1 pt]

Correctly references result from (b) $\rightarrow$ 0.5 mark

Correctly applies equivalence from (a) $\rightarrow$ 0.5 mark

Part (d): Right Inverse to Invertible [1 pt]

Constructs solution $X=CY$ or equivalent idea $\rightarrow$ 0.5 mark

Correctly concludes invertibility using part (a) $\rightarrow$ 0.5 mark

(for c and d parts if they are not solved using previous answers, 0.25 mark each are cut).

Question No 5 (5 Marks) a)-2m, b)-1m, c)-2m

Q5 LA subik.
a) $\vec{0} = \vec{0} + \vec{0}$ (additive identity)
multiply both sides by scalar c

$$c\vec{0} = c(\vec{0} + \vec{0})$$

$$c\vec{0} = c\vec{0} + c\vec{0} \quad (\text{distributivity})$$

now since $c\vec{0}$ is a vector

$\therefore -c\vec{0}$ exists.

adding $-c\vec{0}$ both sides

$$-c\vec{0} + c\vec{0} = c\vec{0} + c\vec{0} - c\vec{0}$$

$$\vec{0} = c\vec{0} + \vec{0} \quad (c\vec{0} - c\vec{0} = \vec{0})$$

$$\vec{0} = c\vec{0} \quad (\vec{0} + \vec{0} = \vec{0})$$

$$\boxed{\vec{0} = c\vec{0}} \quad \text{HP}$$

b) $0 = 0 + 0$ (multiply by vector $\vec{x}$ both sides)

$$0\vec{x} = (0+0)\vec{x}$$

$$0\vec{x} = 0\vec{x} + 0\vec{x} \quad (\text{distributivity scalar addition})$$

Add $-0\vec{x}$ both sides

$$\vec{0} = 0\vec{x} + \vec{0}$$

$$\vec{0} = 0\vec{x} \quad (\vec{x} + \vec{0} = \vec{x})$$

HP

c) as $c \neq 0$, c^{-1} exists.

$$c\vec{x} = \vec{0}$$

multiply both sides by c^{-1}

$$c^{-1}(c\vec{x}) = c^{-1}(c\vec{0})$$

$$(c^{-1}c)\vec{x} = c^{-1}\vec{0}$$

$$1\vec{x} = c^{-1}\vec{0}$$

from part a

$$c\vec{0} = \vec{0}$$

where c is scalar

here c^{-1} is scalar

$$\therefore 1\vec{x} = \vec{0} \quad \therefore \vec{x} = \vec{0} \quad \text{hence proved}$$

Question No 6 (5 Marks)

Proof. Let W be the subspace spanned by S . Then each linear combination

$$\alpha = x_1\alpha_1 + x_2\alpha_2 + \cdots + x_m\alpha_m$$

of vectors $\alpha_1, \alpha_2, \dots, \alpha_m$ in S is clearly in W . Thus W contains the set L of all linear combinations of vectors in S . The set L , on the other hand, contains S and is non-empty. If α, β belong to L then α is a linear combination,

$$\alpha = x_1\alpha_1 + x_2\alpha_2 + \cdots + x_m\alpha_m$$

of vectors α_i in S , and β is a linear combination,

$$\beta = y_1\beta_1 + y_2\beta_2 + \cdots + y_n\beta_n$$

of vectors β_j in S . For each scalar c ,

$$c\alpha + \beta = \sum_{i=1}^m (cx_i)\alpha_i + \sum_{j=1}^n y_j\beta_j.$$

Hence $c\alpha + \beta$ belongs to L . Thus L is a subspace of V .

Now we have shown that L is a subspace of V which contains S , and also that any subspace which contains S contains L . It follows that L is the intersection of all subspaces containing S , i.e., that L is the subspace spanned by the set S . ■

- Parts a, b - 1.5 for correct proof, 0.5 - 1 for intuition and explanation without full proof.
- Part c - 1 for showing L is subspace, 1 for showing L contains S . 0.5 - 1 for intuition and explanation without full proof.